SOLUTION REQUIREMENTS – LegalEase

Date: 29/09/2026 | TeamID:SWTID-2026-5113 | ProjectName:LegalEaseAI | MaximumMarks: 4 Marks

Functional Requirements (FR) – What the system should do

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Users should securely access LegalEase and provide required identity/user information before protected document-generation features. Sensitive API credentials must be stored securely in environment variables.	High
2	Authorization levels	Normal users can enter legal-document details, generate, edit, preview and download documents; administrative/developer access can manage application configuration and API credentials.	Medium
3	External interfaces	Integrate Streamlit frontend with FastAPI through the /generate API. Connect the backend to Google Gemini for AI-powered generation. Support DOCX, PDF and TXT generation modules.	High
4	Transactions processing	Process requests containing document type, parties, terms and effective dates. Validate using Pydantic DocumentRequest, send structured input to Gemini, receive generated content and return it to the frontend.	High
5	Reporting	Provide generated-document preview and export outputs in TXT, DOCX and PDF. DOCX/PDF outputs should preserve structured formatting, headings, terms tables/bullets, branding and footer information.	High
6	Business rules, etc.	Generate documents according to selected document type and supplied parties, terms and dates. Terms may be semicolon-separated clauses. Users must be able to edit generated content before downloading.	High
7	Compliance to laws or regulations	Protect sensitive user and API data, use secure environment-variable configuration for the Gemini API key, and handle generated legal content through controlled application flows.	High
8	Other	Provide responsive Streamlit UI with inputs, Generate Document and Edit Document actions, styled HTML preview, logo/branding support, and download options for TXT, DOCX and PDF.	Medium

Non-Functional Requirements (NFR) – How the system should perform

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Provide responsive document-generation, preview and export workflows. Backend requests should be handled efficiently and the UI should remain responsive during normal use.	Interactive response under normal load; no unnecessary UI blocking.
2	Scalability	Support scaling the FastAPI backend and Streamlit frontend independently, with Gemini handling AI generation and modular document-formatting utilities.	Components can be deployed separately; dependencies/configuration managed through requirements and environment setup.
3	Security & Data Privacy	Protect sensitive user data and Gemini API credentials. API keys must not be hard-coded and should be stored in .env/environment configuration.	No API key in source code; secure environment configuration used.
4	Reliability & Availability	Provide a working FastAPI backend, Streamlit frontend, Gemini integration and document-export pipeline. Verify end-to-end functionality during testing.	Health/root check and successful generate, preview, edit and TXT/DOCX/PDF export tests.
5	Usability & Accessibility	Provide clear inputs for document type, parties, terms and dates; styled preview; edit option; and straightforward download controls.	User can complete generate → preview → edit → download without programming knowledge.
6	Maintainability & Portability	Use modular routes, validators, formatting utilities and configurable dependencies. Support local deployment and documented cloud deployment options.	Modular structure; requirements.txt; deployable in supported environments.

Source basis: supplied LegalEase project document and Solution Requirements template.